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Quantitative trait mapping in a diallel cross of recombinant inbred lines

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Received: 5 January 2005 / Accepted: 10 February 2005

Abstract

A recombinant inbred intercross (RIX) is created by generating diallel F_1 progeny from one or more panels of recombinant inbred (RI) strains. This design was originally introduced to extend the power of small RI panels for the confirmation of quantitative trait loci (QTL) provisionally detected in a parental RI set. For example, the set of 13 C×B (C57BL/6ByJ×BALB/cByJ) RI strains can, in principle, be supplemented with 156 isogenic F_1s. We describe and test a method of analysis, based on a linear mixed model, that accounts for the correlation structure of RIX populations. This model suggests a novel permutation algorithm that is needed to obtain appropriate threshold values for genome-wide scans of an RIX population. Despite the combinational multiplication of unique genotypes that can be generated using an RIX design, the effective sample size of the RIX population is limited by the number of progenitor RI genomes that are combined. When using small RI panels such as the C × B there appears to be only modest advantage of the RIX design when compared with the original RI panel for detecting QTLs with additive effects. The RIX, however, does have an inherent ability to detect dominance effects, and, unlike RI strains, the RIX progeny are genetically reproducible but are not fully inbred, providing somewhat more natural genetic context. We suggest a breeding strategy, the balanced partial RIX, that balances the advantage of RI and RIX designs. This involves the use of a partial RIX population derived from a large RI panel in which the available information is maximized by minimizing correlations among RIX progeny.

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Introduction

The construction of a recombinant inbred (RI) strain panel typically begins with an outcross between two fully inbred progenitor strains. F_1 hybrid animals are bred to each other to produce F_2 animals, each with a unique genome type. A single pair of F_2 animals is chosen and these mice and their progeny are brother–sister mated for greater than 20 generations to create a new inbred strain. This process is repeated starting with several distinct pairs of F_2 animals to generate a panel of recombinant inbred strains (Bailey 1971). RI strain panels offer some compelling advantages for mapping complex genetic traits compared with intercross (F_2) or backcross (BC) populations, and RI panels have been used to map both Mendelian and quantitative traits in the mouse (Plomin and McClearn 1993; Taylor 1996) and in other organisms (Nuzhdin et al. van Swinderen et al. 1997; Limami et al. 2002). The accumulation of recombination events during inbreeding results in an average fourfold increase in recombination in an RI strain compared with a single-generation genetic map (Haldane and Waddington 1931). RI strains need to be genotyped only once. Multiple animals with the same genotype can be phenotyped. Thus, we can assess phenotypes in multiple environments, measure multiple invasive phenotypes, and can repeat noisy measurements to increase their precision (Knapp and Bridges 1990; Belknap 1998). RI strains are homozygous at essentially all loci, which can increase power for detecting additive quantitative trait loci (QTL) effects compared with F_2 populations in which half of animals are heterozygous at any given locus.

There are several drawbacks to working with RI lines as well. RI strains are fully inbred and thus it

 DOI: 10.1007/s00335-004-2466-1 • Volume 16, 344–355 (2005) •

is not possible to estimate dominance effects using RI strains. Moreover, the natural buffering effects of heterozygosity will be absent in RI strains (Hartwell 2004). The inbreeding phase of RI strain generation in mice can take 4–5 years, and newly generated strains require dense genotyping. Thus, the development of a RI strain panel requires considerable effort and cost. This high initial barrier has restricted the development of mouse RI panels and the resulting small size of most panels limits their usefulness, especially for quantitative trait analysis (Taylor 1996; Peirce et al. 2004; Williams et al. 2001, 2004). The power to detect a QTL increases with increasing number of RI strains (Belknap et al. 1996; Belknap 1998). Thus, the limited number of distinct recombinant genomes is the most serious disadvantage of mapping with RI strain panels compared with other mapping populations that can be rapidly expanded to hundreds or thousands of individuals.

A common feature of any small RI panel is the occurrence of nonsyntenic associations in which pairs of loci on different chromosomes have similar or identical strain distribution patterns (SDPs) and therefore appear to be linked. This association can arise either as a result of random fixation of alleles on different chromosomes during the production of RI strains or as a result of selection for particular combinations of alleles on different chromosomes. Unlinked loci showing allelic association can lead to false-positive signals when mapping complex traits (Williams et al. 2001). For example, if a QTL is linked to one marker from an associated pair, a high LOD score will also be generated at the other marker. In the case of perfect allelic association, the LOD scores would be identical and there is no analytic approach that can resolve them. In contrast, segregating populations (F_2 or BC) can generate a large number of distinct genotypes, and, for mapping populations greater than 100 animals, nonsyntenic associations of this type are very rare events.

The recombinant inbred intercross (RIX) strategy was first introduced by Threadgill et al (2002) as a means to dramatically increase the number of unique recombinant genomes available using an RI panel. RIX animals are the F_1 progeny from a mating between two different RI strains. A panel of n RI strains can generate $n \times (n-1)$ unique RIX genotypes, and, furthermore, no additional genotyping is required since the RIX genotype can be inferred from the parental RI genotypes. RIX animals are reproducible and they share many of the other advantages of an RI panel. RIX animals are heterozygous at one half of all loci which will allow estimation of dominance effects. RIX animals are not fully inbred and thus represent a more realistic genetic constitution. Phenotypic variances within F_1 may be lower than for inbred RI of animals because of the buffering capacity of their hemi-heterozygous genome structure.

There are disadvantages to mapping with an RIX cross as well. As we demonstrate here, the combinatorial increase of unique RIX genomes does not overcome the limitation of having a small RI panel. RIX genomes retain the same recombinations as the parental RI strains. There are no new recombinations and moreover, the pattern of nonsyntenic association in the RI set will be retained in the RIX cross.

Analysis of the RIX mapping population presents some challenges. RIX progeny are related to one another in different degrees. Widely used statistical methods for mapping quantitative trait loci (QTL) assume that all individuals are equally correlated and thus effectively independent. The correlation structure of RIX progeny violates this assumption. Here we propose a mixed-linear-model analysis (Littell et al. 1996) and a novel permutation test to account for the correlation among RIX progeny. We present a comparison between the mixed-model approach and standard QTL analysis methods using brain weight phenotype from a C $\times$ B (C57BL/6ByJ $\times$ BALB/cByJ)-derived RIX population. We then use simulations to develop further insights into the statistical nature of the RIX cross. We provide some suggestions for more effectively using the RIX design that are particularly applicable to larger RI panels such as the L $\times$ S set (n = 77) (Williams et al. 2004), the enlarged B $\times$ D set (n = 80) (Peirce et al. 2004), or the proposed Collaborative Cross (n = 1000) (Churchill et al. 2004).

Material and methods

Diallel cross. An RIX cross has the same genetic structure as a classical diallel cross (Hayman 1954a, Griffing 1956) in which the progeny of all pairwise and reciprocal matings are generated from a set of inbred lines. Suppose a set of p inbred lines is chosen and crosses among these lines are made. This procedure gives rise to a maximum of p^2 unique combinations of haploid genomes. This number is doubled when both sexes are considered. Data from such combinations can be set out in a $p \times p$ table in which x_{ii} represents the mean phenotype of the ith parental inbred strain, x_{ij} represents the mean phenotype of the F_1 hybrid from crossing the ith and jth inbred strains, and x_{ji} represents the mean for the reciprocal cross. A diallel cross is full (or complete) if it includes all the $p(p-1)$ crosses and p parental lines (Table 1A); otherwise, it is partial (or incomplete). A half-diallel cross is a partial diallel with only one member of each reciprocal cross pair. The half-diallel cross may or may not include the parental lines (Table 1B). A balanced partial-diallel cross can be formed by selecting a subset of crosses in which each parental strain occurs an equal number of times as both the mother and the father (Table 1C). For example, a balanced partial diallel of 16 hybrids can be derived from a set of 8 parental lines by choosing each RI line twice as the mother and twice as the father in a set of matings. Progeny families belonging to the same row or column in the diallel array have one parent in common, the progeny families in the same row or column are half-sibs. The relationship among RIX progeny can be divided into three categories: (1) RIX animals that share no common parents, and (2) RIX animals that share exactly one common parent, and (3) reciprocal pairs that share two parents in common.

The simplest statistical model of a diallel cross excludes crosses within parental types (Table 1B) and assumes that there are no sex-linked effects and no mitochondrial, cytoplasmic, or epigenetic parental effects. The latter assumptions imply that the reciprocal crosses will share the same mean trait value so we could collapse the data to a half diallel (Griffing 1956; Wright 1985). This model for the phenotype of the kth offspring of the $i \times j$ mating can be written as

$$Y_{ijk} = \mu + g_i + g_j + s_{ij} + \epsilon_{ijk} \qquad (1)$$

where μ is the population mean, g_i and g_j are the general combining abilities (GCA) of parents i and j, s_{ji} is the specific combining abilities (SCA) of $i \times j$ matings, and ϵ_{ijk} is random error associated with kth individual within the $i \times j$ progeny. In this model the values of the mother and father effects are tied together as indicated in the notation g. In other words, the GCA of a strain is the same regardless of whether the strain was the mother or the father of the F_1 progeny. GCA describes the average performance of a parent in hybrid combination with other genotypes. SCA describes the degree to which specific parental combinations lead to deviations in progeny phenotypes from expectations based on the average parental performance. When the available data are strain averages, the index k is dropped and SCA terms are absorbed into the error. Classical diallel analysis does not use genetic marker data or address the problem of mapping the genetic loci responsible for phenotypic variation.

Table 1A. A full-diallel cross from 8 inbred lines

1	1×2	1×3	1×4	1×5	1×6	1×7	1×8
2×1	2	2×3	2×4	2×5	2×6	2×7	2×8
3×1	3×2	3	3×4	3×5	3×6	3×7	3×8
4×1	4×2	4×3	4	4×5	4×6	4×7	4×8
5×1	5×2	5×3	5×4	5	5×6	5×7	5×8
6×1	6×2	6×3	6×4	6×5	6	6×7	6×8
7×1	7×2	7×3	7×4	7×5	7×6	7	7×8
8×1	8×2	8×3	8×4	8×5	7×8	8×8	8

Table 1B. Half-diallel cross from 8 inbred lines without selfing

	1×2	1×3	1×4	1×5	1×6	1×7	1×8
		2×3	2×4	2×5	2×6	2×7	2×8
			3×4	3×5	3×6	3×7	3×8
				4×5	4×6	4×7	4×8
					5×6	5×7	5×8
						6×7	6×8
							7×8

Table 1C. Balanced partial-diallel cross from 8 inbred lines

	1×2			1×5			
			2×4			2×7	
				3×5			3×8
4×1		4×3					
	5×2				5×6		
6×1							6×8
		7×3			7×6		
			8×4			8×8	

Simple model. Linear-model-based analysis of quantitative phenotypes has proven to be a powerful and robust method for QTL mapping in classical populations derived from biparental crosses between inbred lines (Lander and Botstein 1989). The observed phenotype y_i for the ith individual with QTL genotype $m(i)$ in a sample of size n may be described by the linear model

$$y_i = \mu + Q_{m(i)} + \epsilon_i \qquad (2)$$

where μ is the population mean, $Q_{m(i)}$ is the effect of the mth QTL genotype, and ϵ_i is the residual error, usually assumed to be distributed as $N(0, \sigma_\epsilon{}^2)$. Note that $Q_{m(i)}$ may represent a multiple locus genotype. Interaction terms or covariates, such as gender, can also be incorporated in the above model (Ahmadiyeh et al. 2003). We refer to this model as the *simple model*. It assumes that all individuals are equally correlated and thus effectively independent. Therefore, it is suitable for backcross, intercross, and unreplicated RI populations. In the case of replicated RI populations, model (2) can be applied to strain averages. However, in the case of RIX populations, model (2) is not appropriate. The strains are related in various degrees and a mixed-model analysis is needed to account for their correlation structure.

Mixed model. We have adapted the half-diallel model (1) to include the effects of genetic loci in an RIX population. We define a linear mixed model in which the GCA (additive effects) of parents are random effects and the QTL genotypes are fixed effects. The SCA effects are assumed to be negligible here and have been absorbed in the error component. The observed phenotype y_{ijk} of the kth individual with mother i and father j can be written as

$$y_{ijk} = \mu + g_i + g_j + Q_{m(i,j)} + \epsilon_{ijk} \quad (3)$$

where μ, g_i and g_j are as in model (1) and $m(i,j)$ denotes the genotype class at the QTL from progeny of the $i \times j$ cross. By varying the genomic positions of putative QTLs in this model, we can carry out genome scans to map genetic loci using the RIX design. We refer to model (3) as the *half-diallel model* (HD).

An alternative model that does not assume that the mother and father effects are tied together and allows σ_M^2 and σ_P^2 to be different can be written as

$$y_{ijk} = \mu + M_i + P_j + Q_{m(i,j)} + \epsilon_{ijk} \quad (4)$$

where μ and $Q_{m(i,j)}$ are as in model (3). M_i and P_j are the additive effects of the ith mother and jth father, respectively. M_i, P_j, and ϵ_{ijk} are random effects, distributed as N(0, σ_M^2), N(0, σ_P^2), and N (0, σ_ϵ^2), respectively. We refer to model (4) as the *Maternal–Paternal model* (MP model). Both models HD and MP can be applied to strain average data by removing index k from the equation.

Further extensions of models (3) and (4) could be considered to include effects that are specific to reciprocal hybrids, such as parent of origin or extranuclear effect, by including additional terms as described by Griffing (1956). However, one important difference between a standard diallel cross and an RIX is the partitioning of variance generated by Y chromosomes and the mitochondrial genome. These do not produce variance in an RIX. The only difference between 7 × 2 and 2 × 7 is in the parental nongenetic, nonmitochondrial contribution. It is also possible to entertain variations on the RIX/diallel design itself. For example, if two different RI panels are available, the RIX could include crosses between panels.

Implementation. In general, a mixed linear model can be expressed in matrix form as $\mathbf{y} = \mathbf{Xb} + \mathbf{Zu} + \epsilon$, where $\mathbf{y}$ is the vector of observations, $\mathbf{X}$ and $\mathbf{Z}$ are the known design matrices, $\mathbf{b}$ is the unknown vector of fixed effects, and $\mathbf{u}$ is the unknown vector of random effects. In our case the GCA effects and ϵ are the random terms and are assumed to be normally distributed with mean

$$E\begin{bmatrix} U \\ \varepsilon \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and variance

$$\mathrm{Var}\begin{bmatrix} U \\ \varepsilon \end{bmatrix} = \begin{bmatrix} G & 0 \\ 0 & R \end{bmatrix}.$$

Thus, $E(\mathbf{y}) = \mathbf{Xb}$ and $\mathrm{Var}(\mathbf{y}) = \mathbf{ZGZ} \pm \mathbf{R}$. In most genetic analyses, we make the additional assumption that $\mathbf{G}$ is a diagonal matrix and $R = \sigma^2\boldsymbol{I}$. We use this approach to construct a likelihood ratio test statistic for the QTL effects.

For both HD and MP models, the columns of $\mathbf{X}$ (fixed effects) include mean terms and additional columns representing the QTL genotypes. For the HD model, the $\mathbf{Z}$ matrix (random effects) has one column for each parental strain, and for the MP model, $\mathbf{Z}$ has two columns for each parental strain representing different maternal and paternal contributions.

Although standard software is available for fitting linear mixed models, the structure of the HD model requires a special construction for the $\mathbf{Z}$ matrix. SAS software (SAS Institute, Cary, NC) can be used to implement the mixed model for QTL mapping. For example, one could write an SAS macro to fit the HD model in SAS by combining PROC MIXED with PROC IML procedures. Dummy variables can be constructed for each parent using PROC IML as described in the Appendix of Xiang and Li (2001). The combined RIX marker data and dummy variables can then be analyzed with PROC MIXED (Littell et al. 1996) with marker data declared as fixed effect in the MODEL statement and dummy variables for parents in multiple RANDOM statements. The option "TYPE=TOEP(1)" must be used to constrain the same variance over dummy variables in half diallel design. Alternatively, one can use the "TYPE=LIN" covariance structure in the RANDOM statement of PROC MIXED to model covariances between different types of relatives in the RIX as linear combinations of unknown. An example of SAS code of how to construct the coefficient matrices data set associated with the TYPE=LIN(q) option for a diallel design has been given in a newly published SAS book (Fry 2004).

We have implemented a set of functions in the R computing environment, a freely available statistical package (http://www.r-project.org/), to fit these models and to carry out genome scan analyses. The R functions used for this article are available at http://www.jax.org/staff/churchill/labsite/index.html under the software link. The R environment provides a variety of tools that can be used for further analysis and manipulation of the data, including function available in the R/qtl package (Broman et al. 2003).

Permutation tests. It is common practice in QTL studies to control type I error for multiple testing due to genome-wide searching using family-wise error rate control (Westfall and Young 1993). Permutation analysis provides a method to compute multiple test-adjusted thresholds or adjusted p values (Churchill, and Doerge 1994). We typically permute the phenotype data and keep genotype data intact. The maximum logarithm of the odds ratio (LOD) score is recorded on each permutation and percentiles of their distribution provide approximate multiple test-adjusted thresholds.

The significance threshold for the likelihood ratio test statistic from the mixed model is based on a unique type of permutation test. Simple shuffling of the RIX progeny phenotypes does not conserve the correlation structure of the mapping population, and, if this procedure is applied, it will result in inappropriately low thresholds. In general, failure to account for positive correlation in data will lead to overly liberal thresholds. In a sense the data have been overly randomized and the effective information content of the permutated data is greater than in the original data. The extremely liberal thresholds obtained by naively permuting RIX data are illustrated in the analysis and simulations below. However, we note that the parental RI genotypes are exchangeable under the null hypothesis and we can construct a permutation of the genotypes by shuffling the ''labels'' on the parental RI strains and then reconstructing the RIX progeny genotypes based on the shuffled RI strains. For example, consider a panel of RIX progeny derived from a set of 8 parental RI strains. The original labels of the RI strains are [1, 2, 3, 4, 5, 6, 7, 8]. After permutation, the shuffled labels changed to [4, 1, 2, 8, 6, 7, 3, 5]. The permuted genotype for $RIX_{2\times7}^{\star}$ will be reconstructed from $RI_2^{\star}$ and $RI_7^{\star}$, which correspond to the RI_1 and RI_3 before shuffling. We run genome scan on the reconstructed RIX genotypes and record the maximum LOD score and use the percentiles of this distribution to establish a genome-wide adjusted significance threshold. This ''RIX shuffle'' retains the correlation structure of the actual RIX progeny and provides unbiased significance thresholds. In application below we used 1000 shuffles and computed 95th percentile thresholds. However, in simulations we used 200 shuffles and computed 90th percentile thresholds because of the computational demand of permuting within a simulation.

Results

Analysis of the C × B data. The C × B RI panel consists of 13 strains established by Don Bailey and colleagues about 30 years ago at the Jackson Laboratory (Dux et al. 1978). Our C × B RIX mapping population consists of 78 nonreciprocal C × B RIX hybrids, a complete half-diallel cross of the 13 parental C × B RI strains. Twelve mice were generated (roughly 6 male and 6 female) for each RIX hybrid, and total brain weight was measured as described (Williams 2000). Significant effects of age, body weight, and gender on brain weight were found in C × B RIX animals. Therefore, individual RIX brain weight was adjusted for these factors prior to computing strain mean after adjustment, the heritability of brain weight in RIX is 38.8%. Mean adjusted brain weight was used in QTL analysis. With the half-diallel design of the C × B RIX population, we cannot directly confirm that the reciprocal matings will produce the same phenotype distribution. The marginal mean (haplome averages) of brain weight for the RIX hybrids and their 13 parental lines are presented in Table 2. Two-sample t-tests indicate that these are consistent with marginal homogeneity predicted by the HD model. If marginal homogeneity did not hold, the MP model would be a more appropriate choice here. We note that in this half-diallel design, strains do not appear as maternal and paternal contributors to the RIX in equal proportions, and two strains (C × BRI_1 and C × BRI_{13}) contribute to the RIX in only one direction.

C × B RIX genotype data is imputed from parental C × B RI genotypes, which are available at http://www.nervenet.org/papers/bxn.html. The C × B RI genetic map has a cumulative length of approximately 4959 cM, representing roughly 3.6-fold expansion relative to an intercross. A total of 598 markers were typed on C × B RI set and the average resolution spacing is 1–2 cM. There are 374 unique observed SDPs in the C × B RI marker genotypes. The set of RIX strains has a genotype distribution pattern that resembles an F_2 intercross with a 1:2:1 ratio of CC, CB, and BB genotypes at most loci.

We conducted genome-wide scans using the simple, HD, and MP models to identify QTL associated with brain weight at 5-cM increments (in RI map units) over the entire genome. Significance thresholds were determined by permutation analysis

Table 2. Marginal brain weight of RIX hybrids and their parental RI lines

C × B RI set			*C × B RIX hybrid*						
Strain	*N*	*Mean*	*Maternal RI strain*	*N*	*Marginal mean*	*Paternal RI strain*	*N*	*Marginal mean*	*p value*
1	28	454.17	1	12	489.17	1	0	—	—
2	40	438.86	2	11	467.40	2	1	476.43	—
3	62	475.34	3	10	497.01	3	2	485.05	0.39
4	36	453.63	4	9	489.12	4	3	481.21	0.50
5	27	432.30	5	8	468.56	5	4	479.96	0.18
6	66	460.75	6	7	472.71	6	5	488.27	0.11
7	90	480.68	7	6	485.20	7	6	480.88	0.57
8	43	463.53	8	5	480.18	8	7	483.17	0.77
9	47	473.80	9	4	475.49	9	8	487.52	0.13
10	53	446.83	10	3	465.35	10	9	487.45	0.11
11	52	426.12	11	2	464.01	11	10	463.29	0.95
12	49	458.03	12	1	468.15	12	11	486.49	—
13	62	421.00	13	0	—	13	12	471.42	—

using simple shuffling and RIX shuffling. Thus, we examined a total of six combinations of models and shuffling methods. Results are summarized in Table 3 and in Fig. 1.

Results from the C × B RIX data showed that shuffling the RIX progeny phenotype generated inappropriately low threshold values. Using the simple model [2] 13 QTL peaks rise above the significance threshold based on the simple shuffle, whereas no QTL reached the significance threshold based on the RIX shuffle method (Fig. 1A). Both mixed models, HD and MP, show a single peak on Chromosome 11 that just exceeds the significance threshold based on the RIX permutation method (Fig. 1B). In Table 3 we show that all 12 additional peaks detected by the simple model (with simple permutation) have a high nonsyntenic correlation with the Chromosome 11 locus, strongly suggesting that these are false positives generated by spurious association with the Chromosome 11 QTL. This occurrence of false positives in the simple model scan is a compelling reason to use the mixed model for RIX analysis.

Table 3. Significant QTL peaks in the C × B RIX analyses

	LOD scores		
Chr No.	*Simple model*[a]	*MP model*[b]	*Correlation with Chr 11 locus*
1	5.8	—	0.59
3	9	—	–0.59
5	6.6	—	0.43
8	9.5	—	0.69
9	7.2	—	0.43
10	8.1	—	–0.59
11	10.8	19.3	1.0
12	5.5	—	0.57
13	5.3	—	–0.51
14	6.0	—	–0.51
15	7.9	—	0.59
17	5.3	—	0.43
18	7.4	—	0.59

[a]QTL peaks above significant threshold (LOD score = 4.0, $p < 0.05$) from simple shuffle + simple QTL. None of the peaks here above the significant threshold (LOD = 16.4) based on RIX shuffling method.
[b]QTL peaks above significant threshold (LOD score = 19.3, $p < 0.05$) from RIX shuffling method + MP QTL model.

Analysis of a simulated phenotype on the C × B RIX genotypes

Simulation 1.1: QTL on Chromosome 5. We simulated a hypothetical phenotype in the RIX population using the C × B genotype data, assuming a QTL with moderate effect on Chromosomes 5 is present. The results of genome scans using the simple model and the mixed models are presented in Fig. 2. All three models failed to detect a significant peak on Chromosome 5 when the RIX shuffle threshold value is used. Failure to detect the simulated QTL on Chromosome 5 by either of the mixed-model analyses (using the RIX shuffle threshold) is most likely a consequence of the small size of the RI panel used to generate the RIX population resulting in low statistical power. Table 4 summarizes the QTL peaks above the threshold value based on the simple shuffling method. Repeated simulations showed that the simple model had a higher false-positive rate

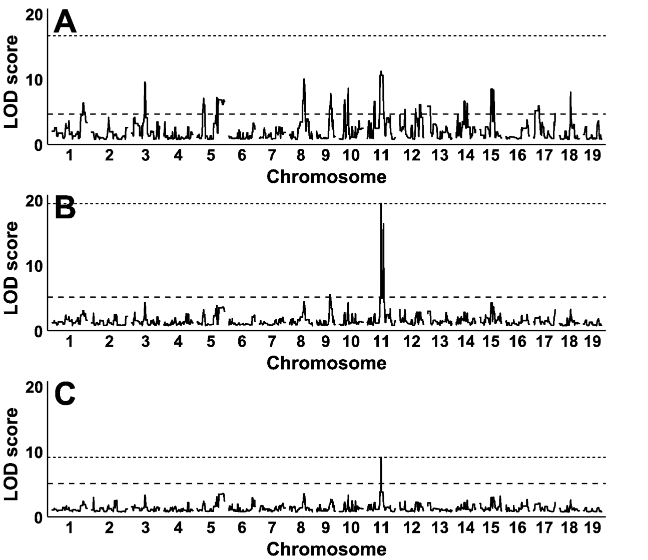

Fig. 1. Genome-wide scan of adjusted mean brain weight from the C × B RIX population. LOD scores were computed using the simple QTL model (**A**), the MP model (**B**), and the HD model (**C**). Genome-wide significance ($p < 0.05$) based on 1000 permutations using the simple shuffle (dashed line) and the RIX shuffle (dotted line) methods.

Table 4. Simulated QTL peak on Chromosome 5[a]

	LOD scores			
Chr No.	*Simple model*[b]	*MP model*[c]	*HD model*[d]	*Correlation with Chr 5*
2	6.9	—	12.7	−0.63
5	12.3	10.7	14.5	1.00
6	9.0	4.8	5.0	−0.85
9	5.1	—	—	−0.63
10	4.3	—	—	0.68
12	9.0	4.9	8.7	0.84
13	5.3	—	—	−0.54
14	6.9	—	—	0.85
16	4.6	—	—	0.54
19	4.3	—	—	−0.50

[a]QTL peaks exceeding the simple shuffling threshold.
[b]Significance thresholds for $p < 0.10$ are 3.7 (simple shuffle) and 12.5 (RIX shuffle).
[c]Significance thresholds for $p < 0.10$ are 4.2 (simple shuffle) and 10.7 (RIX shuffle).
[d]Significance threshold for $p < 0.10$ are 4.3 (simple shuffle) and 16.2 (RIX shuffle).

than either mixed model, regardless of the shuffling method and that simple shuffling thresholds resulted in an excess of false-positive detections for all three models. In the example simulation (Table 4), all of the false-positive QTLs are highly correlated with the simulated QTL on Chromosome 5.

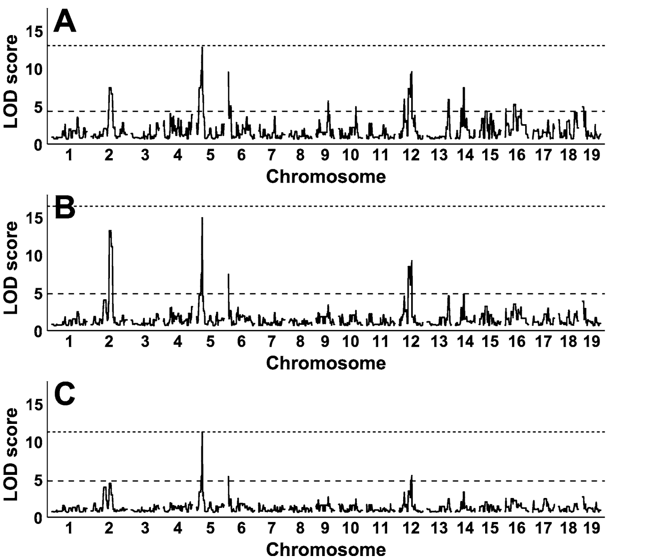

Fig. 2. Genome-wide scan of a hypothetical phenotype. Data were simulated based on the C × B genetic marker data with a major QTL (effect size 1.5) on Chromosome 5. Normal residual error was simulated with standard deviation 1.0. LOD scores were computed using the simple QTL model (**A**), the MP model (**B**), and the HD model (**C**). Genome-wide significance ($p < 0.1$) based on 200 permutations using the simple shuffle (dashed line) and the RIX shuffle (dotted line) methods.

Simulation 1.2: A ghost QTL. When the number of strains in an RI panel is small, the occurrence of nonsyntenic association will be high. Mapping with an RIX population generated from a small RI panel presents the same problem. A total of 8 pairs of unlinked loci (typed markers) show perfect association in the C × B RIX genotype data. For example, a typed marker on Chromosome 10 has a perfect association with another typed marker on Chromosome 3. We simulated a hypothetical phenotype on the RIX population using the C × B genotype data with a QTL on Chromosome 10 to illustrate the problem of nonsyntenic association in RIX. A genome scan using the MP model showed a ghost QTL on Chromosome 3 with the same LOD score and effect size as the QTL on Chromosome 10 (Fig. 3). When the allelic association is perfect, as in this example, there is no analytic approach to resolve the ghost QTL that arise by nonsyntenic associations. If the nonsyntenic association is imperfect, it may be possible to identify the causal locus by conditioning on each locus and assessing the conditional LOD at

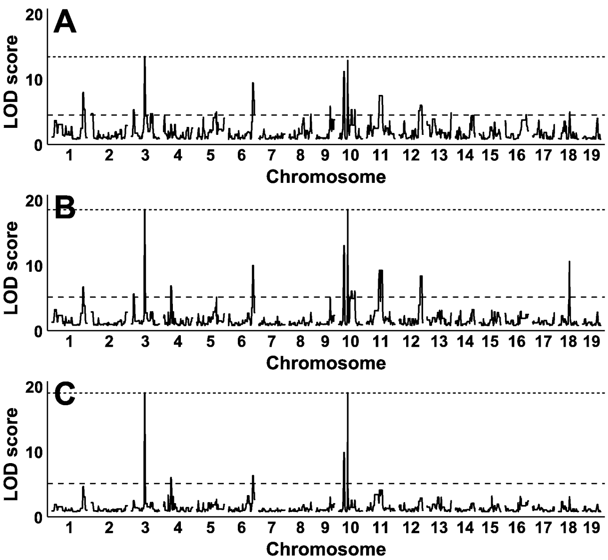

Fig. 3. Ghost QTL simulation. Data were simulated based on the C × B genetic marker data with a major QTL (effect size 1.5) on Chromosome 10 to illustrate the problem of nonsyntenic association in RIX. Normal residual error was simulated with standard deviation 1.0. LOD scores were computed using the simple QTL model (**A**), the MP model (**B**), and the HD model (**C**). Genome-wide significance ($p <$ 0.1) based on 200 permutations using the simple shuffle (dashed line) and the RIX shuffle (dotted line) methods.

the others. However, this will result in substantial loss of power. Moreover, nonsyntenic association can affect many loci and, if there are multiple QTLs, the cumulative effects can create many spurious LOD peaks. The best solution to nonsyntenic association may be to avoid the problem altogether by using a sufficiently large panel of RI strains to establish the RIX or to use a partial RIX.

Simulation 2: RI simulation. We conducted simulations to assess the effect of the size of an RI panel on (1) the occurrence of nonsyntetic association, (2) the power to detect QTL, and (3) the size and distribution of permutation-based threshold values.

Simulation 2.1. The first simulation demonstrates that nonsyntetic association is expected to occur at high frequency in small RI panels and diminishes as the panel size is increased. We simulated RI strain panels of sizes 16, 32, 64, 128, 256, 1000, 5000, and 10,000. We repeated each simulation 100 times and recorded the maximum absolute correlation coefficient among unlinked loci. The distribution of maximum correlation is shown in Fig. 4A. We conclude that an RI panel of at least 128 strains is needed to ensure that the maximum correlation between unlinked loci is below 0.4. This simulation can not account for biologically driven nonsyntenic associations generated (perhaps) by

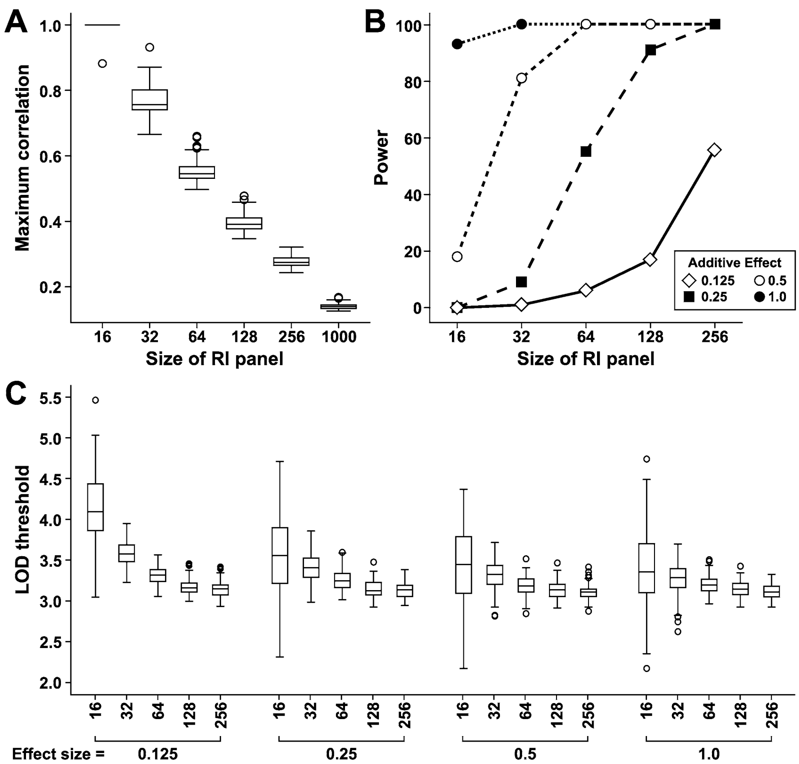

Fig. 4. RI line simulations. Distribution of maximum correlation among unlinked markers from RI panels with different number of strains in 100 simulations (**A**). Power to detect the major QTL effect across RI panels with different number of strains. Each RI panel has a phenotype determined by three background QTLs (effect size 0.125) on Chromosomes 1, 10, and 16, and one major QTL on Chromosome 5 with effect size ranging from 0.125 to 1.0 (**B**). Distribution of the significance thresholds for LOD scores at the 0.90 genome-wide adjusted level. Threshold values were determined using 200 permutations within each RI simulation (**C**).

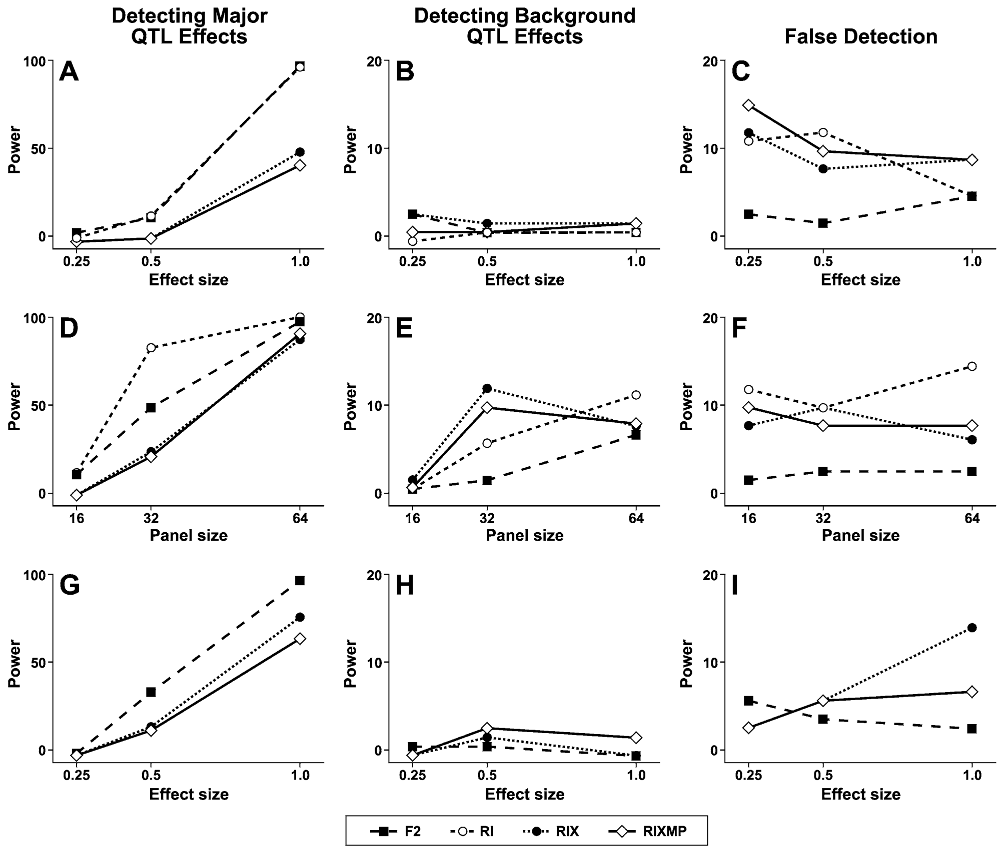

Fig. 5. Power comparison of a balanced partial RIX mapping population with an RI panel and an F_2 using populations of the same size, in total number of animals. The top panel summarizes the comparison of an RI panel with 16 strains, and F_2 and RIX population of size 64 while the major QTL effect size varied from 0.25 to 1.0 on Chromosome 5 (**A**–**C**). The middle panel summarizes the power comparison of RI, F_2, and RIX when major QTL effect size is fixed at 0.5 across different sizes of the mapping panels (**D**–**F**). The bottom panel summarizes the power of detecting dominant QTL effect size ranging from 0.25 to 1 between a partial RIX cross and an F_2 population of size 64 (**G**–**I**).

epistatic interactions that affect fitness; thus, 128 is an ''idealized'' minimum and it is likely that higher strain numbers would be needed to dilute nonsyntenic association in real populations.

Simulation 2.2. The second RI-based simulation compares the power for detecting a QTL with additive effects across RI panels with different numbers of strains. We simulated RI panels with 16, 32, 64,128, and 256 strains. Each RI strain has a phenotype determined by one major QTL on Chromosome 5 and three background QTLs on Chromosomes 1, 10, and 16. Four possible effect sizes (1.0, 0.5, 0.25, and 0.125) are used for the major QTL and the background QTL effects are fixed at 0.125. We simulated the hypothetical phenotype four times and analyzed the strain means. Significance thresholds at the 0.90 adjusted level were determined using 200 permutations within each simulation. We repeated each combination 100 times and recorded the number of times that we detected the major QTLs as well as detection rates for background QTLs and false positives. Results are summarized in Fig. 4B. Power to detect the major QTL depends on both the number of RI strains and the effect size of the major QTL relative to the background QTLs. For example, if the effect size of the major QTL is 0.25, an RI panel with 128 strains has 90% power to detect it. Power is reduced to less than 60% if the RI panel is reduced to 64 strains.

The distributions of threshold values obtained from these simulations are shown as box plots in Fig. 4C. It is clear that the mean and the variance of the threshold depend on both the size of the RI sets and the effect size of the major QTL. When the number of RI strains is small and when the major QTL effect is large, threshold values are very unstable. This necessitates the use of permutation for each simulation replicate.

Comparison of RIX, RI, and F_2

Simulation 3. QTL with additive effects We conducted a set of simulations to compare the power of an RIX mapping population with an RI panel and an F_2 using populations of the same size, in total number of animals. We simulated RI panels with N = 16, 32, and 64 strains, a balanced partial RIX cross of 64, 128, and 256 hybrids from the RI strains, and an F_2 cross with 64, 128, and 256 mice. Each RI line occurs four times as the mother and four times as the father of an RIX hybrid. We simulated one major QTL with additive effect on Chromosome 5 and three additive background QTLs on Chromosomes 1, 10, and 16. Four effect sizes (1.0, 0.5, 0.25, and 0.125) are used for the major QTL and the background QTL effects are fixed at 0.125. We simulated the phenotype four times and ran genome scan on the mean trait value in each RI strain. The RIX and F_2 populations were simulated with QTLs in the same locations with the same additive QTL effects as in the RI but individuals were not replicated. In all cases the same total number of individual animals was simulated. We analyzed data using the simple model and the MP mixed model. We repeated the simulations 100 times to compare the power of detecting QTL on Chromosome 5 with respect to the size and type of the cross and the effect size of QTLs.

The top panel of Fig. 5 (A–C) summarizes the simulation results for an RI panel with 16 strains and F_2 and RIX populations of size 64. The middle panel of Fig. 5(D–F) summarizes the simulation results when the major QTL effect is fixed at 0.5 and the size of the RI panel varies from 16 to 64. When the sample size is small and the major QTL has an effect size close to that of the background QTLs, none of the crosses had sufficient power to detect this additive QTL (Fig. 5A). As expected, we observed a higher false-positive rate in RI and RIX compared with that in F_2 (Fig. 5C). This is because the extent of nonsyntenic association is higher in the RI and RIX designs than in the F_2 design. Consistent with our finding in simulation 2.1, the false-positive rate goes down in RI and RIX when the size of the RI panel increases (Fig. 5F). Although this simulation shows that RIX has less power to detect an additive QTL than RI and F_2, the power of the RIX improves when the size of RI panels increases (Fig. 5D). Finally, the results of the ranges of our simulated models indicate that, with the RIX shuffling method, the simple model is no worse than the MP model at detecting the major QTL on Chromosome 5. They also indicate that the MP model may have better power in detecting major QTL when the number of parental RI strains is increased.

Simulation 4: QTL with dominant effects. We simulated an RI panel of 16 strains and used it to generate RIX populations with dominant QTL effects using the same strategy as in simulation 3. Results are summarized in the bottom panel of Fig. 5 (G–I). When a small panel of RI is used, RIX has limited power to detect the dominant QTL compared with an F_2 population of the same size. We also note that the simple model has slightly better power for detecting the major QTL than does the MP model. However, the simple model has a higher false-positive rate when the effect of the QTL increases and the high rate of false-positive detections is a cause for concern.

Conclusion

The RIX mapping design was introduced to improve the power of small RI panels for the detection of QTLs. The RIX design shares some of the best and some of the least desirable properties of small parental RI panels. We describe a method of analysis, based on a linear mixed model, and a method of permuting the data to obtain valid thresholds for genome-wide scans of an RIX population. These methods account for the correlation structure of the RIX population. Despite the combinational explosion of unique genotypes that are generated in an RIX design, there appears to be little advantage of the RIX compared with the original RI panel for detecting QTLs with additive effects. The effective sample size of the RIX population is limited by the number of progenitor RI strains used to establish the cross. Nevertheless, there are reasons to consider RIX as a viable genetic mapping design. The ability to study traits in the context of a genetic background with a low formal inbreeding coefficient is one potentially valuable feature of the RIX design. This is difficult to assess with simple simulation studies and awaits further experience with RIX animals. It is clear that large RI sets are needed to achieve reasonable power with the RIX design. Nonsyntenic associations arise when regions on different Chromosomes show statistically significant positive or negative correlations. An RI panel with at least 128 strains is needed to reduce the maximum absolute correlation among unlinked loci to 0.4. As illustrated in our examples, nonsyntenic associations greater than 0.4 can result in false-positive QTL results. Practical constraints on sample sizes will require partial-diallel populations like the balanced designs used in our simulations. Another advantage of the partial-diallel design is that it decreases the extent of positive correlation among RIX animals and, thus, the effective sample size approaches the actual sample size.

RI and RIX designs both support studies that incorporate multiple genetic, environmental, and developmental variables into comprehensive models of disease susceptibility. With the combinatorial diversity available in the vast number of potential RIX genomes that can be generated from a large set of RI strains, it will be possible to construct mice to validate statistical models relating genotypes and environments to phenotypic outcomes. With levels of heterozygosity and admixture comparable to human populations, RIX animals represent genetically normal mammals that can be replicated *ad infinitum*.

The mixed model provides the most appropriate statistical description of the RIX population. In our simulations, it did not significantly outperform the simple model in power to detect QTL. However, the mixed model, in combination with the RIX shuffle permutation, substantially reduced the frequency of false positives resulting from nonsyntenic linkage. The mixed model correctly takes into account the correlation structure of the RIX population. The result of this correction is a reduction in the information content (effective sample size) of the RIX population compared with an equivalent number of independently sampled animals. The balanced partial-RIX design (Table 1C), in which each parental RI strain is used an equal number of times as mother and as father in generating the RIX progeny, provides a potential solution to the problem of correlation among the RIX progeny. If the founder RI panel is sufficiently large, a sparse sampling of the RIX population can achieve near independence. Therefore, we recommend the use of partial RIX populations derived from large RI panels for future applications of this breeding design in QTL mapping applications.

Acknowledgments

This work was supported by the National Institutes of Health (NIH) grants HL55001 and GM070683 and NIMH, NIDA and NSFI grant P20-MH62009. Additional support for this work was provided by the Department of the Army, Medical Division, Gant DAMD-LRD. The authors thank Dr. David Threadgill for discussion of the RIX design.

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